PROJECT LOGIC

Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Platform	ServiceNow
Core Objective	Control access to records using the user's roles and the Branch field value.
Scope	READ, CREATE, WRITE and DELETE ACLs for the Institution Details table (u_institution_details).

1. Project Objective

This project shows how script-controlled Access Control Lists (ACLs) in ServiceNow can limit access to records based on a user's roles and on the value stored in a record. EEE users are allowed to see EEE branch records, and the administrator keeps full access. Three more roles are then used to control who can create, edit and delete records.

2. Project Logic / Functional Flow

1. Create the EEE test user and the custom roles bb1, bb2, bb3 and bb4.
2. Create the custom table Institution Details (u_institution_details).
3. Add the fields for student roll number, student name, faculty name, branch, email, phone number and description.
4. Add sample records with the Branch values ECE, EEE and CSE.
5. Create a READ ACL for the table that uses the condition Branch = EEE and an advanced script.
6. In the script, allow administrators and users with bb1, and deny everyone else.
7. Create the CREATE ACL (bb2), the WRITE ACL (bb3) and the DELETE ACL (bb4).
8. Test the permissions by impersonating users and opening the table list.
9. Check that each role combination can do only the operations it is meant to do on the permitted EEE records.

3. Data and Security Objects

Object	Configuration	Purpose	Expected Result
User	EEE User	Used to test role-based access	The user gets the custom roles that are assigned
Roles	bb1, bb2, bb3, bb4	One role for each operation	Roles can be combined for a user
Table	u_institution_details	Stores the institution and student records	Records are protected by the ACLs
Branch field	ECE / EEE / CSE	Value used for the record-level condition	READ is limited to EEE records
ACL – READ	bb1 + Branch = EEE + script	Controls who can view records	bb1 users see EEE records; admin sees all
ACL – CREATE	bb2	Controls who can add records	bb2 is required
ACL – WRITE	bb3	Controls who can edit records	bb3 is required
ACL – DELETE	bb4	Controls who can delete records	bb4 is required

4. READ ACL Decision Logic

The advanced script of the READ ACL is:

```
(function () {
    if (gs.hasRole('admin')) {
        return true;
    }

    if (gs.hasRole('bb1')) {
        return true;
    }

    return false;
})();
```

The READ ACL also has the data condition Branch is EEE. So the READ behaviour is: administrator → full access; bb1 user → EEE records; user without the required role → no records.

5. Permission Matrix

User / Role	READ	CREATE	WRITE	DELETE	Checked By
Administrator	All records	All records	All records	All records	Administrator access check
bb1	EEE records	No	No	No	READ ACL test
bb1 + bb2	EEE records	Yes	No	No	CREATE ACL test
bb1 + bb2 + bb3	EEE records	Yes	Yes	No	WRITE ACL test
bb1 + bb2 + bb3 + bb4	EEE records	Yes	Yes	Yes	DELETE ACL test
No required role	No records	No	No	No	Negative test for READ

6. Verification Logic

- Impersonate the EEE user and open the Institution Details list.
- Check that only the EEE branch records are shown.
- Impersonate a user without the required role and check that no records are shown.
- Impersonate the administrator and check that records of all branches are shown.
- Add bb2 to the test user and check that the New button is available for creating a record.
- Add bb3 and check that an EEE record can be edited.
- Add bb4 and check that an EEE record can be deleted.
- Open the list during testing with u_institution_details_list.do.

7. End-to-End Project Logic

User identity → Assigned roles → ACL operation → Branch condition → Script evaluation → Allow / Deny decision → List and form behaviour

- The user identity tells ServiceNow who is asking for access.
- The assigned roles decide whether the user is eligible for the operation.
- The ACL is picked according to the operation: read, create, write or delete.
- For READ, the Branch condition limits the records to EEE.
- The advanced script checks for the administrator and bb1.
- ServiceNow then allows or denies the request, and the list or form shows the result.

8. Project Outcome

The project helps us understand how script-controlled READ ACLs and separate CREATE, WRITE and DELETE ACLs work in ServiceNow. It shows how a user's roles and the field value of a record are checked together, and how ACLs protect data from being viewed, changed, created or deleted by users who are not allowed to do so.